



Lista 2 - Cálculo II

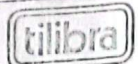
① a) $(1+i) - (2-3i)$ $|z| = \sqrt{1^2 + 2^2}$ $\bar{z} = a - bi$
 $(1-2) + (1+3i)$ $\rightarrow |z| = \sqrt{1+16}$ $\rightarrow \bar{z} = -1 - 2i$
 $-1 + 4i$ $|z| = \sqrt{17}$ $\rightarrow -1 - 2i$

b) $(4-11/2i) - (9+5/2i)$ $|z| = \sqrt{5^2 + 3^2}$ $\bar{z} = a - bi$
 $(4-9) + (-11/2 - 5/2)i$ $\rightarrow |z| = \sqrt{25+9}$ $\rightarrow \bar{z} = -5 - (-3i)$
 $(-5) + (-6/2i)$ $|z| = \sqrt{36}$ $\bar{z} = -5 + 3i$
 $-5 - 3i$ $|z| = 6$ $\rightarrow -5 + 3i$

b) $(2-7i) \cdot (1+3i)$ $|z| = \sqrt{25^2 + 5^2}$ $\bar{z} = a - bi$
 $(2+12i-7i-21i^2)$ $|z| = \sqrt{625+25}$ $\rightarrow \bar{z} = 25 - 5i$
 $-21 + 2 + 5i$ $\rightarrow |z| = \sqrt{650}$ $25 - 5i$
 $25 + 5i$

d) $\frac{5-i}{3+4i} \cdot \frac{3-4i}{3-4i}$ $|z| = \sqrt{\frac{11^2 - 23i^2}{25}}$ $\bar{z} = a - bi$
 $\frac{15-20i-3i+4i^2}{9-12i+12i-16i^2}$ $\rightarrow |z| = \sqrt{\frac{650}{625}}$ $\rightarrow \bar{z} = \frac{11}{25} + \frac{23i}{25}$
 $\frac{-11-23i}{25}$ $|z| = \sqrt{\frac{26}{25}}$ $\rightarrow \bar{z} = \frac{11}{25} + \frac{23i}{25}$
 $\frac{11}{25} - \frac{23i}{25}$

e) $\frac{3}{2i-3i}$ $|z| = \sqrt{\frac{12}{25} + 9}$ $\bar{z} = a - bi$
 $\frac{3}{2i-3i}$ $|z| = \sqrt{\frac{12+81}{625}}$ $\bar{z} = \frac{12}{25} - \frac{9i}{25}$
 $\frac{3}{2i-3i}$ $|z| = \sqrt{\frac{225}{625}}$ $\rightarrow \bar{z} = \frac{12}{25} - \frac{9i}{25}$
 $\frac{12+9i}{16+12i-12i-9i^2}$ $|z| = \frac{15}{25}$
 $\frac{12+9i}{25}$ $|z| = \frac{3}{5}$
 $\frac{12}{25} + \frac{9i}{25}$



$$\textcircled{1} 1 + i^{101}$$

$$1 + (i^4)^{25} \cdot i$$

$$1 + (1)^{25} \cdot i$$

$$1 + (1 \cdot 1) \cdot i$$

$$1 + i$$

$$|z| = \sqrt{1^2 + 1^2}$$

$$|z| = \sqrt{1 + 1}$$

$$|z| = \sqrt{0}$$

$$|z| = 0$$

$$|z| = a - bi$$

$$|z| = 1 - i$$

$$1 - i$$

$$\textcircled{2} a) (3 + 4i)^2 - 2\bar{z} = z$$

$$(3 + 4i) \cdot (3 + 4i) - 2\bar{z} = z$$

$$9 + 12i + 12i + 16i^2 - 2\bar{z} = z$$

$$-7 + 24i - 2\bar{z} = z$$

$$-7 + 24i = z + 2\bar{z}$$

$$-7 + 24i = (a + bi) + 2(a - bi)$$

$$-7 + 24i = (a + bi) + 2a - 2bi$$

$$-7 + 24i = (3a - bi)$$

$$-7 = 3a \quad 24i = -bi$$

$$-7 = 3a \quad -(-24) = 24$$

$$\frac{-7}{3} = a \quad \frac{24}{2} = 12$$

$$b) 1z + 3\bar{z} = 5 - 2i$$

$$1(a + bi) + 3(a - bi) = 5 - 2i$$

$$a + bi + 3a - 3bi = 5 - 2i$$

$$4a - 2bi = 5 - 2i$$

$$(4a - b) = 5$$

$$(2b) = 2$$

$$4a - b = 5$$

$$2b = 2 \quad 8b = 11$$

$$a - 3b = -2 \quad \times 3$$

$$-3a + 9b = 6 \quad b = 11$$

$$3a - 11/8 = 5 \rightarrow 3a = 40 + 11$$

$$24a = 51$$

$$a = 51/24$$

$$\frac{17}{8} + \frac{11}{8}i$$

$$c) \left(\frac{1+i}{1-i} \right)^2$$

$$\frac{1+i}{1-i} \cdot \frac{1+i}{1-i}$$

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$$\frac{1+i}{1-i} \cdot \frac{1+i}{1-i}$$

tilibra

tilibra

12)

$$z^2 = 4 + 2i\sqrt{5}$$

$$(a+bi) \cdot (a+bi) = 4 + 2i\sqrt{5}$$

$$a^2 + 2abi + b^2 i^2 = 4 + 2i\sqrt{5}$$

$$a^2 - b^2 + 2abi = 4 + 2i\sqrt{5}$$

$$a^2 - b^2 = 4$$

$$a^2 - b^2 = 4$$

$$2abi = 2\sqrt{5}i$$

$$a \cdot b = \sqrt{5} \rightarrow b = \frac{\sqrt{5}}{a}$$

$$x^2 = 5 \quad x = \pm 5 \quad x^2 = -1 \quad x = \pm i$$

$$(a^2 - 5) = 4 \quad \times a^2$$

$$a^4 - 5 = 4a^2 \rightarrow a^4 - 4a^2 - 5 = 0$$

$$x^2 = 4x - 5 = 0$$

$$5 \text{ or } -1$$

9)

$$a) 9z^2 + 16 = 0$$

$$9z^2 = -16$$

$$z^2 = -16/9$$

$$z = \frac{\sqrt{-16}}{3} \rightarrow z = \frac{-4i}{3}$$

c)

$$2z^2 - 2z + 1$$

$$1 + 2i \text{ or } 1 - 2i$$

$$+2 \pm \sqrt{2 \cdot 2 \cdot 1}$$

$$+2 \pm \sqrt{-4}$$

$$\frac{2 \pm 4i}{2}$$

b)

$$z^4 - 1 = 0$$

$$z^4 = x$$

$$x - 1 = 0$$

$$\frac{-0 \pm \sqrt{0^2 - 1 \cdot 1}}{2} \rightarrow +\frac{2}{2} \text{ or } -\frac{2}{2} + 1 \text{ or } -1$$

d)

$$z^2 + z + 2$$

$$\frac{-1 \pm \sqrt{1-8}}{2}$$

$$-1 \pm \sqrt{1-8}$$

$$-1 \pm \sqrt{-7}$$

$$-1 \pm \sqrt{7}i$$

$$\frac{-1 \pm \sqrt{7}i}{2}$$

e)

$$z^4 + 3z^2 + 2 = 0$$

$$z^2 = y$$

$$y^2 + 3y + 2$$

$$-3 \pm \sqrt{9-4 \cdot 1}$$

$$\frac{-3+1}{2} \text{ or } \frac{-3-1}{2}$$

$$\rightarrow -1 \text{ or } -2$$

$$z^2 = -1 \rightarrow \pm i$$

$$z^2 = -2 \rightarrow z = \sqrt{-2} \rightarrow z = \pm \sqrt{2}i$$

f)

$$z^4 - 2z^2 + 4 = 0$$

$$z^2 = y$$

$$y^2 - 2y + 4$$

$$+2 \pm \sqrt{2 \cdot 2 \cdot 4}$$

$$+2 \pm \sqrt{-12}$$

$$\frac{2 \pm 2\sqrt{3}i}{2}$$

$$1 \pm \sqrt{3}i$$

$$z^2 = 1 \pm \sqrt{3}i$$

tilibra

4) a) $z + \bar{z} = 2\text{Re}(z)$

$$a+bi + a-bi = 2\text{Re}(z)$$

$$2a = 2\text{Re}(z)$$

b) $z - \bar{z} = 2im(z)$

$$a+bi - (a-bi) = 2im$$

$$a+bi - a+bi = 2im$$

$$2bi = 2im$$

c) $|a| \leq |a+bi|$

$$\sqrt{a^2} \leq \sqrt{a^2+b^2}$$

$$a \leq a+b$$

d) $|z_1+z_2|^2 = |z_1|^2 + |z_2|^2 + 2\text{Re}(z_1\bar{z}_2)$

$$(z_1+z_2) \cdot \overline{(z_1+z_2)}$$

$$(z_1+z_2) \cdot (\bar{z}_1+\bar{z}_2)$$

$$(z_1\bar{z}_1)(z_2\bar{z}_2)(z_2\bar{z}_1)(z_1\bar{z}_2)$$

$$|z_1|^2 + |z_2|^2 + 2\text{Re}(z_1\bar{z}_2)$$

e) $|z_1+z_2| \leq |z_1|+|z_2|$

$$\sqrt{z_1^2+z_2^2} \leq \sqrt{z_1^2} + \sqrt{z_2^2}$$

$$z_1+z_2 \leq |z_1|+|z_2|$$

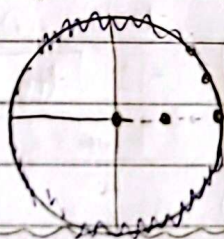
$$a+bi+c+di \leq a+bi+c+di$$

$$a+c+bi+di \leq a+c+bi+di$$

5) a) $|z|=2$

$$(\sqrt{x^2+y^2})^2 = (2)^2$$

$$x^2+y^2 = 4$$



Módulo = 2

Todos valores na extremidade

b) $|z| < 2$

$$(\sqrt{x^2+y^2})^2 < (2)^2$$

$$x^2+y^2 < 4$$

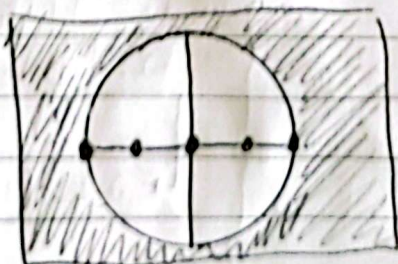


Toda dentro do círculo

c) $|z| > 2$

$$(\sqrt{x^2+y^2})^2 > 2^2$$

$$x^2+y^2 > 4$$



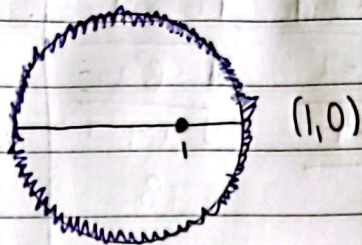
Seria todos fora do círculo.

$$a) |z-1| = 2$$

$$|x+yi-1| = 2$$

$$(\sqrt{x^2+y^2-1^2})^2 = 2^2$$

$$x^2+y^2+1 = 4$$

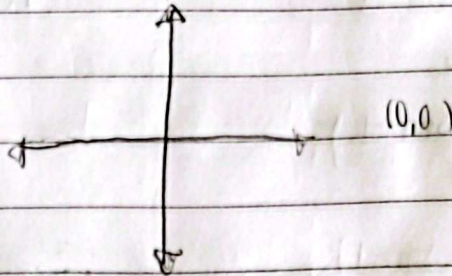


$$e) |z+1| = |z-1|$$

$$|x+yi+1| = |x+yi-1|$$

$$(\sqrt{x^2+y^2+1^2})^2 = (\sqrt{x^2+y^2-1^2})^2$$

$$x^2+y^2+1 = x^2+y^2-1$$



$$f) |z+1| \geq |z|$$

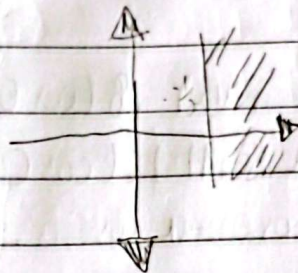
$$|x+yi+1| \geq |x+yi|$$

$$(\sqrt{(x+1)^2+y^2})^2 \geq (\sqrt{x^2+y^2})^2$$

$$x^2+2x+1+y^2 \geq x^2+y^2$$

$$2x \geq -1$$

$$x \geq -1/2$$



$$g) z = r(\cos \theta + i \sin \theta)$$

$$\frac{1}{z} = \frac{1}{r}(\cos \theta + i \sin \theta)$$

$$\frac{1}{z} = \frac{1}{r(\cos \theta + i \sin \theta)}$$

$$\frac{\cos \theta - i \sin \theta}{r(\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta)}$$

$$\frac{\cos \theta - i \sin \theta}{\cos^2 \theta - \cos \theta \sin \theta + \cos \theta \sin \theta - \sin^2 \theta}$$

$$\frac{\cos \theta - i \sin \theta}{\cos^2 \theta - \sin^2 \theta - 1}$$

$$\frac{\cos \theta - i \sin \theta}{r(\cos^2 \theta + \sin^2 \theta)}$$

$$\frac{\cos \theta - i \sin \theta}{r} = \frac{1 \cdot (\cos \theta + i \sin \theta)}{r}$$

$$\frac{11}{6} - \frac{3}{6} \rightarrow \frac{8}{6} \rightarrow \frac{4}{3}$$



Q2

$$\begin{aligned} \text{Modulo} &\rightarrow |2\sqrt{3} - 2i| & \text{Modulo} &\rightarrow |10 + 8i| \\ \sqrt{2\sqrt{3}^2 - 2^2} & & \sqrt{10^2 + 8^2} & \\ \sqrt{12 + 4} & & \sqrt{164} & \\ \sqrt{16} & & 8 & \\ 4 & & & \end{aligned}$$

Argumento z

$$\cos = \frac{a}{p} = \frac{2\sqrt{3}}{4} \rightarrow \frac{\sqrt{3}}{2}$$

Argumento w

$$\cos = \frac{a}{p} = \frac{0}{8} \rightarrow 0$$

$$\text{Sen} = \frac{b}{p} = \frac{-2}{4} \rightarrow -\frac{1}{2}$$

$$\text{Sen} = \frac{b}{p} = \frac{8}{8} \rightarrow 1$$

Argumento é $330 \rightarrow \frac{11\pi}{6}$

Argumento é $\frac{\pi}{2}$

Forma Algebrica

$$Z = 4 \left(\cos \frac{11\pi}{6} + \text{sen} \frac{11\pi}{6} \right) \quad W = 8 \left(\cos \frac{\pi}{2} + \text{sen} \frac{\pi}{2} \right)$$

Z.W

$$4 \cdot 8 \left(\cos \left(\frac{11\pi}{6} + \frac{\pi}{2} \right) + \text{sen} \left(\frac{11\pi}{6} + \frac{\pi}{2} \right) \right)$$

$$32 \left(\cos \frac{7\pi}{2} + \text{sen} \frac{7\pi}{2} \right)$$

$$Z/W = \frac{4}{8} \left(\cos \left(\frac{11\pi}{6} - \frac{\pi}{2} \right) + \text{sen} \left(\frac{11\pi}{6} - \frac{\pi}{2} \right) \right)$$

$$\frac{1}{2} \left(\cos \frac{4\pi}{3} + \text{sen} \frac{4\pi}{3} \right)$$

$$\frac{1}{2} \rightarrow \frac{1}{2} \quad \text{e} \quad -\frac{11}{6} \rightarrow \frac{2\pi - 11}{6} \rightarrow \frac{12\pi - 11\pi}{6} \rightarrow \frac{\pi}{6}$$

$$\hookrightarrow \frac{1}{2} \cdot \left(\cos \frac{\pi}{6} + \text{sen} \frac{\pi}{6} \right)$$

tilibre

Modulo

$$2 \quad \frac{\sqrt{3} + i}{\sqrt{(\sqrt{3})^2 + 1^2}} \quad \frac{1 + i\sqrt{3}}{\sqrt{1^2 + (\sqrt{3})^2}}$$

$$\frac{\sqrt{3} + i}{\sqrt{4}} \quad \frac{1 + i\sqrt{3}}{\sqrt{4}}$$

$$\frac{\sqrt{3} + i}{2} \quad \frac{1 + i\sqrt{3}}{2}$$

Argumento

$$\cos \frac{\alpha}{\rho} = \frac{\sqrt{3}}{2} \quad \cos \frac{\alpha}{\rho} = \frac{1}{2} > \frac{\pi}{3}$$

$$\sin \frac{\beta}{\rho} = \frac{1}{2} \quad \sin \frac{\beta}{\rho} = \frac{\sqrt{3}}{2}$$

Forma Algebraica

$$2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \quad 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

W.2

$$2.2 \left[\cos \left(\frac{\pi}{6} + \frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{6} + \frac{\pi}{3} \right) \right]$$

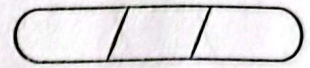
$$2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$$

$$W/2 \quad \frac{2}{2} \left[\cos \left(\frac{\pi}{6} - \frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{6} - \frac{\pi}{3} \right) \right]$$

$$1 \left[\cos -\frac{\pi}{6} + i \sin -\frac{\pi}{6} \right]$$

$$1/2 \quad \frac{1}{2} \quad -\frac{\pi}{6} \rightarrow 2\pi - \frac{\pi}{6} \rightarrow 12 - \pi$$

$$2\pi - \frac{\pi}{2}$$



12. módulo

$$\begin{aligned} & \frac{-\sqrt{3} + 5i}{\sqrt{(-\sqrt{3})^2 + 5^2}} \quad \frac{-3 - 3i}{\sqrt{(-3)^2 + (-3)^2}} \\ & \frac{-\sqrt{3} + 5i}{\sqrt{75 + 25}} \quad \frac{-3 - 3i}{\sqrt{9 + 9}} \\ & \frac{-\sqrt{3} + 5i}{\sqrt{100}} \quad \frac{-3 - 3i}{\sqrt{18}} \rightarrow \sqrt{9 \times 2} \\ & \frac{-\sqrt{3} + 5i}{10} \quad \frac{-3 - 3i}{3\sqrt{2}} \end{aligned}$$

Argumento

$$\begin{aligned} \cos = \frac{a}{p} & \rightarrow \frac{5\sqrt{3}}{10} \rightarrow \frac{\sqrt{3}}{2} \quad \cos \frac{a}{p} = \frac{-3}{3\sqrt{2}} \rightarrow \frac{-1}{\sqrt{2}} \rightarrow \frac{-\sqrt{2}}{2} \\ \sin \frac{b}{p} & \rightarrow \frac{5}{10} \rightarrow \frac{1}{2} \quad \sin \frac{b}{p} = \frac{-3}{3\sqrt{2}} \rightarrow \frac{-\sqrt{2}}{2} \\ & \downarrow 30^\circ \rightarrow \pi/6 \quad 225 \quad 5\pi/4 \end{aligned}$$

Forma Algebraica

$$Z = 10 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \quad W = 3\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right)$$

2.W

$$10 \cdot 3\sqrt{2} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} + \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right)$$

$$30\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$\frac{Z}{W}$

$$\frac{10}{3\sqrt{2}} \left(\cos \frac{\pi}{6} - \frac{5\pi}{4} + i \sin \frac{\pi}{6} - \frac{5\pi}{4} \right) + \frac{6\sqrt{2}}{3} \left(\cos \frac{-13\pi}{12} + i \sin \frac{-13\pi}{12} \right)$$



Modulo

$$\begin{aligned} d) |2\sqrt{3} - 2i| &= |1 - 1 + i| \\ \sqrt{2 \cdot 3 - 2^2} &= \sqrt{-1^2 + 1^2} \\ \sqrt{2 \cdot 3 - 4} &= \sqrt{1+1} \\ \sqrt{6-4} &= \sqrt{2} \\ \sqrt{2} &= \sqrt{2} \end{aligned}$$

Argumento

$$\begin{aligned} \cos = \frac{a}{p} &\rightarrow \frac{2\sqrt{3}}{8} \rightarrow \frac{\sqrt{3}}{2} & \cos = \frac{a}{p} &= \frac{-1}{\sqrt{2}} = \frac{-\sqrt{2}}{2} \\ \sin = \frac{b}{p} &\rightarrow \frac{-2}{8} \rightarrow \frac{-1}{4} & \sin = \frac{b}{p} &= \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \\ &\downarrow & &\downarrow \\ &\frac{1}{6} \rightarrow \frac{11}{6} & &\frac{3\pi}{4} \end{aligned}$$

$$Z = 8 \left(\cos \frac{11\pi}{6} + \sin \frac{11\pi}{6} \right) \quad W = \sqrt{2} \left(\cos \frac{3\pi}{4} + \sin \frac{3\pi}{4} \right)$$

Forma Algebraica

$$8\sqrt{2} \left(\cos \frac{11\pi}{6} + \sin \frac{11\pi}{6} \right)$$

$$8\sqrt{2} \left(\cos \frac{3\pi}{4} + \sin \frac{3\pi}{4} \right)$$

$$\frac{8}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} \rightarrow \frac{8\sqrt{2}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} \rightarrow 4\sqrt{2} \left(\cos \frac{22\pi}{12} - \frac{9\pi}{12} + \sin \frac{22\pi}{12} - \frac{9\pi}{12} \right)$$

$$4\sqrt{2} \left(\cos \frac{13\pi}{12} + \sin \frac{13\pi}{12} \right)$$

	Argumento
a) $(1 - \sqrt{3}i)$ Módulo $ 1 - \sqrt{3}i $ $\sqrt{1^2 + 3}$ $\sqrt{4} \rightarrow 2$	$\cos = a/p \rightarrow \frac{1}{2}$ $\tan = \frac{-\sqrt{3}}{1} \rightarrow \frac{-\pi}{3}$ $\text{Sen} = b/p \rightarrow \frac{-\sqrt{3}}{2}$

Forma Algebraica $2 \cdot (\cos \frac{4\pi}{3} + \text{Sen} \frac{4\pi}{3})$

$$Z^6 = 2^6 (\cos(6 \cdot \frac{4\pi}{3}) + \text{Sen}(6 \cdot \frac{4\pi}{3})) \rightarrow Z^6 = 64 (\cos(8\pi) + \text{Sen}(8\pi))$$

	Argumento
b) Módulo $ 1 - i $ $\sqrt{1^2 + 1^2}$ $\sqrt{2}$	$\cos = a/p \rightarrow 1/\sqrt{2}$ $\tan = \frac{-1}{1} \rightarrow -1 \rightarrow \frac{3\pi}{4}$ $\text{Sen} = b/p \rightarrow -1/\sqrt{2}$

Forma Algebraica $\sqrt{2} (\cos \frac{3\pi}{4} + \text{Sen} \frac{3\pi}{4})$

$$Z^8 = \sqrt{2}^8 (\cos 8 \cdot \frac{3\pi}{4} + \text{Sen} 8 \cdot \frac{3\pi}{4}) \quad Z^8 = 16 (\cos 6\pi + \text{Sen} 6\pi)$$

	Argumento
c) Módulo $ 2\sqrt{3} + 2i $ $\sqrt{2^2 \sqrt{3}^2 + 2^2}$ $\sqrt{16}$ 4	$\cos u/p = \frac{2\sqrt{3}}{4} \rightarrow \sqrt{3}/2 \quad 30^\circ \rightarrow \frac{\pi}{6}$ $\text{Sen} b/p = \frac{2}{4} \rightarrow 1/2$

Forma Algebraica $2^7 \cdot 4^7 (\cos 7 \cdot \frac{\pi}{6} + \text{Sen} 7\pi)$

$$2^7 \cdot 4^7 (\cos \frac{7\pi}{6} + \text{Sen} \frac{7\pi}{6})$$

$$2^7 \cdot 16384 (\cos 7\pi/6 + \text{Sen} 7\pi/6)$$

	Argumento
Módulo $ 1 + i $ $\sqrt{1^2 + 1^2}$ $\sqrt{2}$	$\cos = 1/\sqrt{2} \rightarrow \sqrt{2}/2 \quad 45^\circ \rightarrow \frac{\pi}{4}$ $\text{Sen} = 1/\sqrt{2} \rightarrow \sqrt{2}/2$

Forma Algebraica $\sqrt{2} (\cos \pi/4 + \text{Sen} \pi/4)$

$$\sqrt{2}^{40} (\cos 40 \cdot \frac{\pi}{4} + \text{Sen} 40 \cdot \frac{\pi}{4}) \rightarrow 2^{20} (\cos 10\pi + \text{Sen} 10\pi)$$

Modulo	Argumento	Forma Algebraica
9) a) $10 + i$ $\sqrt{1^2 + 10^2}$ $\sqrt{101}$	$\cos 0/\sqrt{101} \rightarrow 0 \cdot \frac{1}{\sqrt{101}} \rightarrow \frac{\pi}{2}$ $\sin = \frac{1}{\sqrt{101}} \rightarrow \frac{1}{\sqrt{101}} \rightarrow \frac{\pi}{2}$	$10 \left(\cos \frac{\pi}{2} + \sin \frac{\pi}{2} \right)$

$$\sqrt{101} \left(\cos \frac{\pi/2 + 2K\pi}{2} + i \sin \frac{\pi/2 + 2K\pi}{2} \right)$$

$$K=0: \left(\cos \frac{\pi/2 + 0}{2} + i \sin \frac{\pi/2 + 0}{2} \right) \rightarrow \frac{\pi}{4} \quad \sqrt{\frac{2}{2}} \quad \sqrt{\frac{2}{2}}$$

$$\rightarrow 1 \left(\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) \rightarrow \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$K=1 \left(\cos \frac{\pi/2 + 2\pi}{2} + i \sin \frac{\pi/2 + 2\pi}{2} \right)$$

$$1 \left(\cos -\frac{\sqrt{2}}{2} + i \sin -\frac{\sqrt{2}}{2} \right) \rightarrow -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

Modulo	Argumento	Forma Algebraica
$1 + 0i$ $\sqrt{1^2 + 0^2}$ $\sqrt{1}$ 1	$\cos = a/r \rightarrow 1/1 \rightarrow 0$ $\sin = b/r \rightarrow 0/1 \rightarrow 0$	$1 (\cos 0 + \sin 0)$

$$1 (\cos \theta + i \sin \theta) \rightarrow \sqrt[3]{1} \left(\cos \frac{0 + 2 \cdot 0 \cdot \pi}{3} + i \sin \frac{0 + 2 \cdot 0 \cdot \pi}{3} \right) \rightarrow K=0$$

$$1 (\cos 0 + i \sin 0) \rightarrow 1 + 0i$$

$$K=1 \rightarrow 1 \left(\cos \frac{0 + 2 \cdot 1 \cdot \pi}{3} + i \sin \frac{0 + 2 \cdot 1 \cdot \pi}{3} \right) \rightarrow 1 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \rightarrow \frac{1}{2} + \frac{\sqrt{3}}{2} i$$

$$K=2 \rightarrow 1 \left(\cos \frac{0 + 2 \cdot 2 \cdot \pi}{3} + i \sin \frac{0 + 2 \cdot 2 \cdot \pi}{3} \right) \rightarrow 1 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) \rightarrow \frac{1}{2} - \frac{\sqrt{3}}{2} i$$

	Argumento	Forma Algebraica
$ 1+i $	$\cos = a/p = 1/\sqrt{2}$	$\sqrt{2} (\cos \pi/4 + \text{Sen } \pi/4 i)$
$\frac{1^2+1^2}{\sqrt{2}}$	$\text{Sen} = b/p = 1/\sqrt{2}$	
$\sqrt{2}$	$45^\circ \rightarrow \pi/4$	
$\sqrt{2}$		

$$2^{\frac{1}{8}} \left(\cos \frac{\pi/4 + 2 \cdot 0 \pi}{4} + i \sin \frac{\pi/4 + 2 \cdot 0 \pi}{4} \right) \quad k=0$$

$$2^{\frac{1}{8}} \left(\cos \frac{\pi/4 + 2\pi}{16} + i \sin \frac{\pi/4 + 2\pi}{16} \right) \quad \rightarrow$$

$$2^{\frac{1}{8}} \left(\cos \frac{\pi/4 + 2\pi}{4} + i \sin \frac{\pi/4 + 2\pi}{4} \right) \quad k=1$$

$$2^{\frac{1}{8}} \left(\cos \frac{9\pi}{16} + i \sin \frac{9\pi}{16} \right)$$

$$2^{\frac{1}{8}} \left(\cos \frac{\pi/4 + 4\pi}{16} + i \sin \frac{\pi/4 + 4\pi}{16} \right)$$

$$2^{\frac{1}{8}} \left(\cos \frac{17\pi}{16} + i \sin \frac{17\pi}{16} \right) \quad k=2$$

$$2^{\frac{1}{8}} \left(\cos \frac{\pi/4 + 6\pi}{4} + i \sin \frac{\pi/4 + 6\pi}{4} \right) \quad k=3$$

$$2^{\frac{1}{8}} \left(\cos \frac{25\pi}{16} + i \sin \frac{25\pi}{16} \right)$$

d) Módulo	Argumento	Forma Algebraica
$ 0-8i $	$\cos a/p \rightarrow 0/8$	$8 (\cos 3\pi/2 + \text{Sen } 3\pi/2 i)$
$\sqrt{-8^2}$	$\text{Sen } b/p \rightarrow -8/8$	$k=0$
$\sqrt{64}$	$0 \rightarrow -1$	$\sqrt[3]{8} \cdot (\cos \frac{3\pi/2 + 0}{3} + \text{Sen } \frac{3\pi/2 + 0}{3}) \rightarrow 2 (\cos \pi/2 + \text{Sen } \pi/2 i)$
8	$270^\circ \rightarrow 3\pi/2$	$0 + 2i$

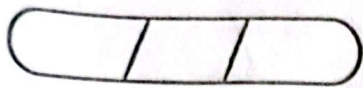
$$k=1$$

$$\sqrt[3]{8} = \left(\cos \frac{3\pi/2 + 2\pi}{3} + \text{Sen } \frac{3\pi/2 + 2\pi}{3} \right) \rightarrow 2 \left(\cos \frac{7\pi}{6} + \text{Sen } \frac{7\pi}{6} i \right) = -\sqrt{3} - i$$

$$\sqrt[3]{8} = \left(\cos \frac{3\pi/2 + 2\pi \cdot 2}{3} + \text{Sen } \frac{3\pi/2 + 2\pi \cdot 2}{3} \right) \rightarrow 2 \left(\cos \frac{11\pi}{6} + \text{Sen } \frac{11\pi}{6} i \right) \quad k=2$$

$$\sqrt{3} - i$$

tilibra



Módulo	Argumento	Forma Algebraica
e) $ -32+0i $ $\sqrt{-32^2}$ $\sqrt{1024}$ $\sqrt{32}$	$\cos = a/r \rightarrow -32/32$ $\sin = b/r \rightarrow 0/32$ $\cos = -1$ y $\sin = 0$ π	$32 (\cos \pi + i \sin \pi)$

$$\sqrt[5]{32} ((\cos \frac{\pi+0}{5}) + i (\sin \frac{\pi+0}{5})) \quad K=0$$

$$2 ((\cos \frac{\pi+2\pi}{5} + i \sin \frac{\pi+2\pi}{5})) \quad K=1$$

$$2 ((\cos \frac{3\pi+4\pi}{5} + i \sin \frac{3\pi+4\pi}{5})) \quad K=2$$

$$2 ((\cos \frac{\pi+6\pi}{5} + i \sin \frac{\pi+6\pi}{5})) \quad K=3$$

$$2 ((\cos \frac{\pi+8\pi}{5} + i \sin \frac{\pi+8\pi}{5})) \quad K=4$$

$$2 ((\cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}))$$

Modulo	Argumento	Forma Algebrica
$64 + 0i$	$\cos = a/p \rightarrow 64/64 \rightarrow 1$	$64 (\cos 0\pi + \text{sen } 0\pi i)$
$\sqrt{64^2}$	$\text{sen} = b/p \rightarrow 0/64 \rightarrow 0$	
64	$\rightarrow 0\pi$	

$$\sqrt[6]{64} = \left(\cos \frac{0+0}{6} + i \sin \frac{0+0}{6} \right) \rightarrow 2 (\cos 0 + i \sin 0) \quad K=0 \text{ (r)} \\ 2 + 0i$$

$$\sqrt[6]{64} = \left(\cos \frac{0+2\pi}{6} + i \sin \frac{0+2\pi}{6} \right) \rightarrow 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \quad K=1. \\ 1 + \sqrt{3}i$$

$$\sqrt[6]{64} = \left(\cos \frac{0+4\pi}{6} + i \sin \frac{0+4\pi}{6} \right) \rightarrow 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \quad K=2. \\ -1 + \sqrt{3}i$$

$$\sqrt[6]{64} = \left(\cos \frac{0+6\pi}{6} + i \sin \frac{0+6\pi}{6} \right) \rightarrow 2 (\cos \pi + i \sin \pi) \quad K=3. \\ -2 + 0i$$

$$\sqrt[6]{64} = \left(\cos \frac{0+8\pi}{6} + i \sin \frac{0+8\pi}{6} \right) \rightarrow 2 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) \quad K=4. \\ -1 - \sqrt{3}i$$

$$\sqrt[6]{64} = \left(\cos \frac{0+10\pi}{6} + i \sin \frac{0+10\pi}{6} \right) \rightarrow 2 \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right) \quad K=5. \\ 1 - \sqrt{3}i$$

$$(10) \quad e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \rightarrow e^{ix} \approx 1 + ix + \frac{(ix)^2}{2} + \frac{(ix)^3}{3}$$

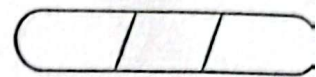
$$\rightarrow e^{ix} \approx 1 + ix - \frac{x^2}{2} - \frac{ix^3}{3} \rightarrow e^{ix} = 1 - \frac{x^2}{2} + i \left(x - \frac{x^3}{3} \right)$$

$\rightarrow$ A questão pediu para verificar se $e^{ix} = \cos x + i \sin x$.

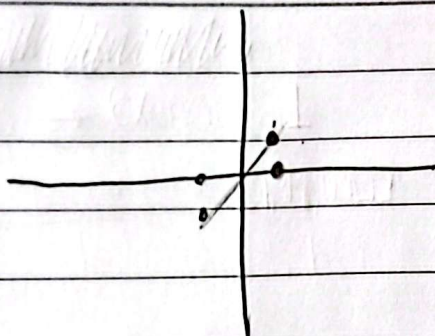
$$\bullet \cos x = 1 - \frac{x^2}{2} \quad \bullet \sin x = x - \frac{x^3}{6}$$

Logo foi provado que se pode escrever

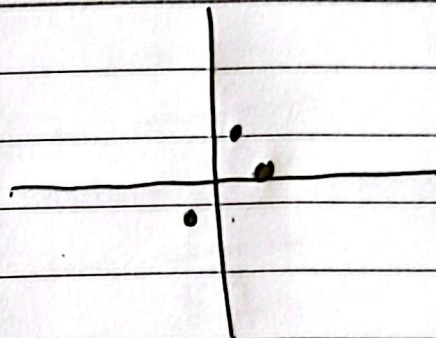
$$\cos \left(1 - \frac{x^2}{2} \right) + i \sin \left(x - \frac{x^3}{3} \right) = \cos \left(1 - \frac{x^2}{2} \right) + i \sin \left(x - \frac{x^3}{3} \right)$$



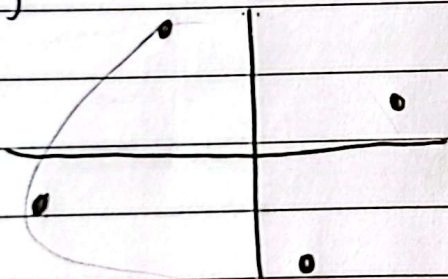
a)



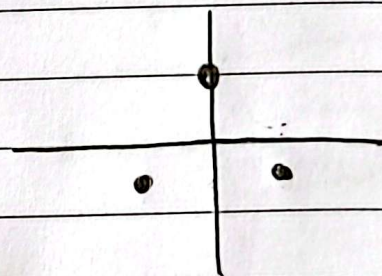
b)



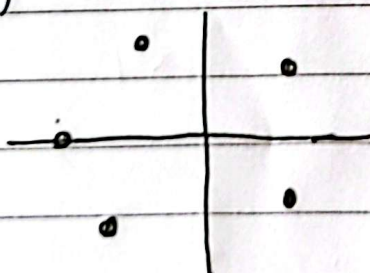
c)



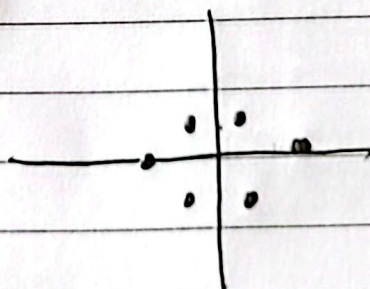
d)



e)



f)



$$e^{i\theta} = \cos \theta + i \sin \theta$$

(1p) a) $e^{-i\pi/2}$

$$\rightarrow \cos \frac{-\pi}{2} + i \sin \frac{-\pi}{2}$$

$$\cos -\pi/2 \rightarrow 0$$

$$\sin -\pi/2 \rightarrow -1$$

Logo na forma retangular é: $(0 - 1i)$

b) $e^{2\pi i}$

$$\rightarrow \cos 2\pi + i \sin 2\pi$$

$$\cos 2\pi \rightarrow 1$$

$$\sin 2\pi \rightarrow 0$$

Logo na forma retangular é: $(1 + 0i)$

c) $e^{i\pi/3}$

$$\rightarrow \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$

$$\cos \pi/3 \rightarrow 1/2$$

$$\sin \pi/3 \rightarrow \sqrt{3}/2$$

Logo na forma retangular é: $(1/2 + i\sqrt{3}/2)$

d) $e^{-i\pi}$

$$\rightarrow \cos -\pi + i \sin -\pi$$

$$\cos -\pi \rightarrow -1$$

$$\sin -\pi \rightarrow 0$$

Logo na forma retangular é: $(-1 + 0i)$

e) $e^{2+i\pi}$

$$\rightarrow \cos \pi + i \sin \pi$$

$$\sin \pi = 0$$

$$\cos \pi = -1$$

$$\rightarrow e^2$$

$$\rightarrow e^2 \cdot -1 \rightarrow -e^2$$

$$\rightarrow (-e^2 + 0i)$$

f) $e^{\pi+i}$

$$\rightarrow \cos \pi + i \sin \pi$$

$$\sin \pi = 0$$

$$\cos \pi = -1$$

$$(-1 + 0i)$$

(12)

$$\left(\frac{e^{ix} - e^{-ix}}{2i} \right)^2 + \left(\frac{e^{ix} + e^{-ix}}{2} \right)^2 = 1$$

$$(\cancel{\cos x} + i \sin x) - (\cancel{\cos x} + i \sin x)$$

$$\frac{2i \sin x}{2i}$$

$$\frac{-(\sin x)^2}{-1}$$

$$\sin^2 x$$

$$(\cancel{\cos x} + i \cancel{\sin x}) + (\cancel{\cos x} - i \cancel{\sin x})$$

$$\frac{2 \cos x}{2}$$

$$(\cos x)^2$$

$$\cos^2 x$$

$$\sin^2 x + \cos^2 x = 1$$